



SOCIAL DEBT

Where Social Science meets Computer Science to bring chaos

Stefano Lambiase



IMAGINE!

Example Scenario

You've spent your life working your way up through the ranks and studying to reach the top.

After years of sacrifice, your boss has finally promoted you.

You are now
MANAGERS!

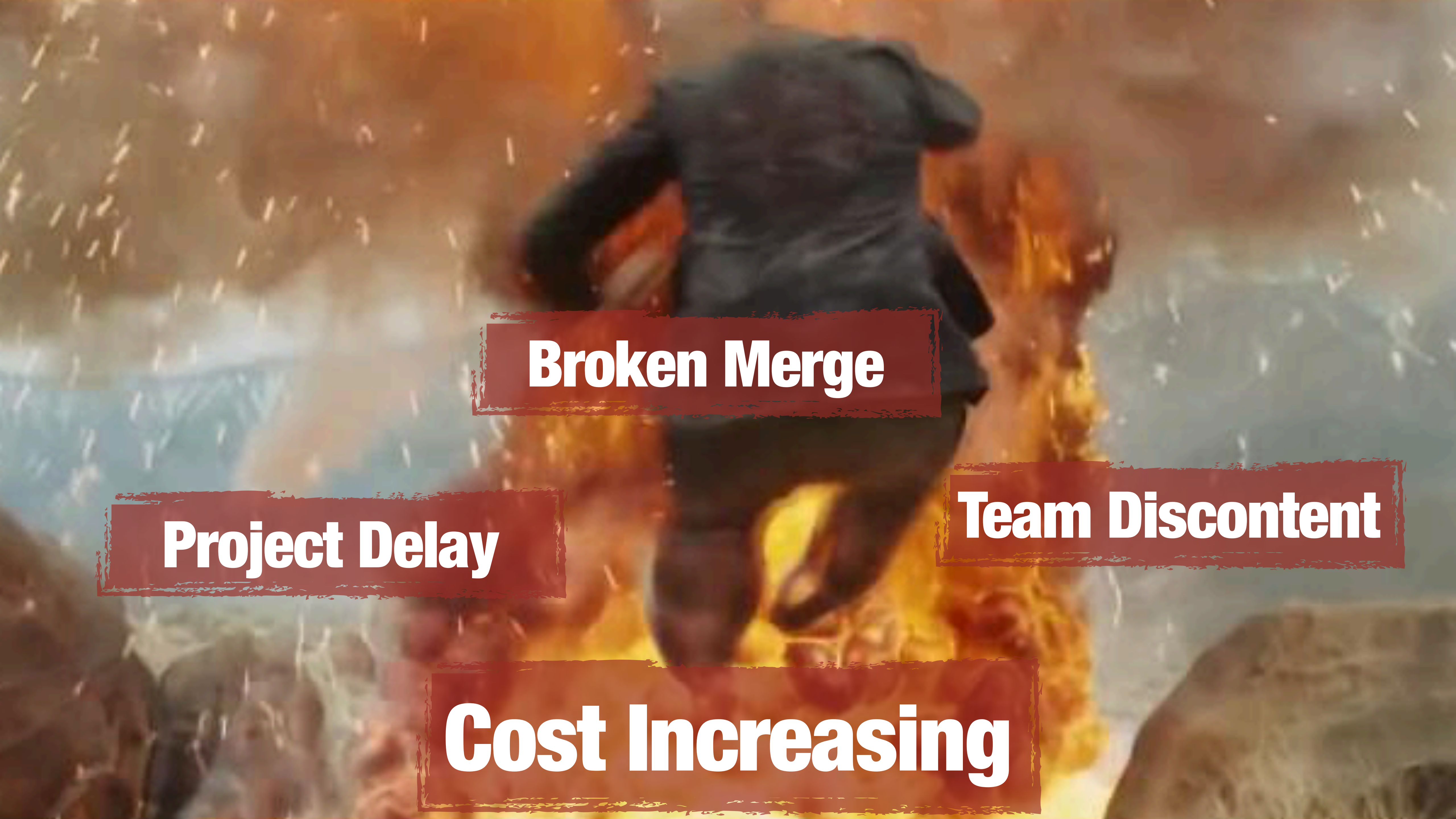
IMAGINE!

Example Scenario

... notwithstanding this, managing human beings (especially engineers) can be difficult, leading to the emergence of unforeseen, and often disruptive, (social) problems.



**Team Member bringing caos
and discontent in your team**



Broken Merge

Project Delay

Team Discontent

Cost Increasing

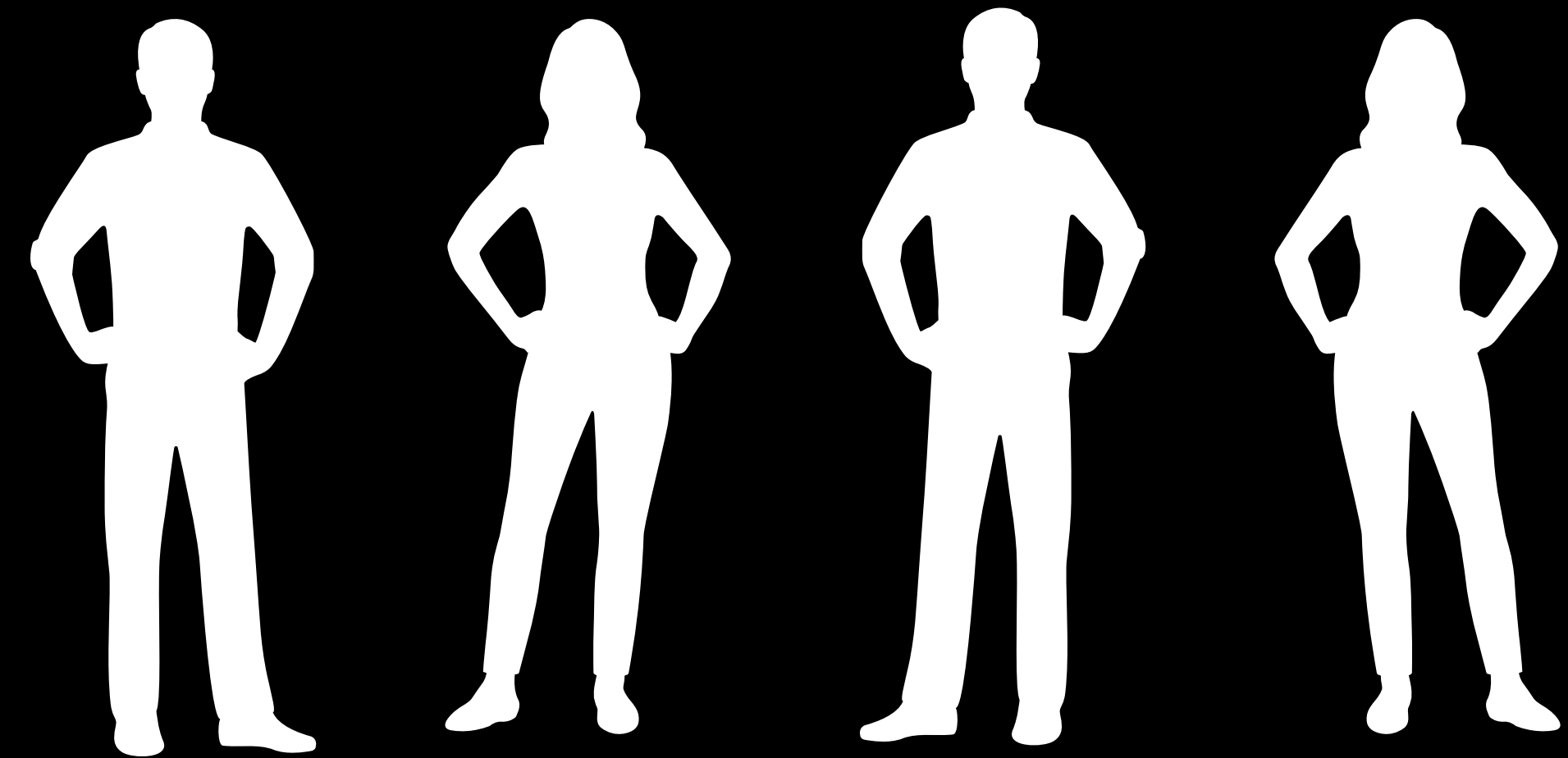


SOCIAL

DEBT

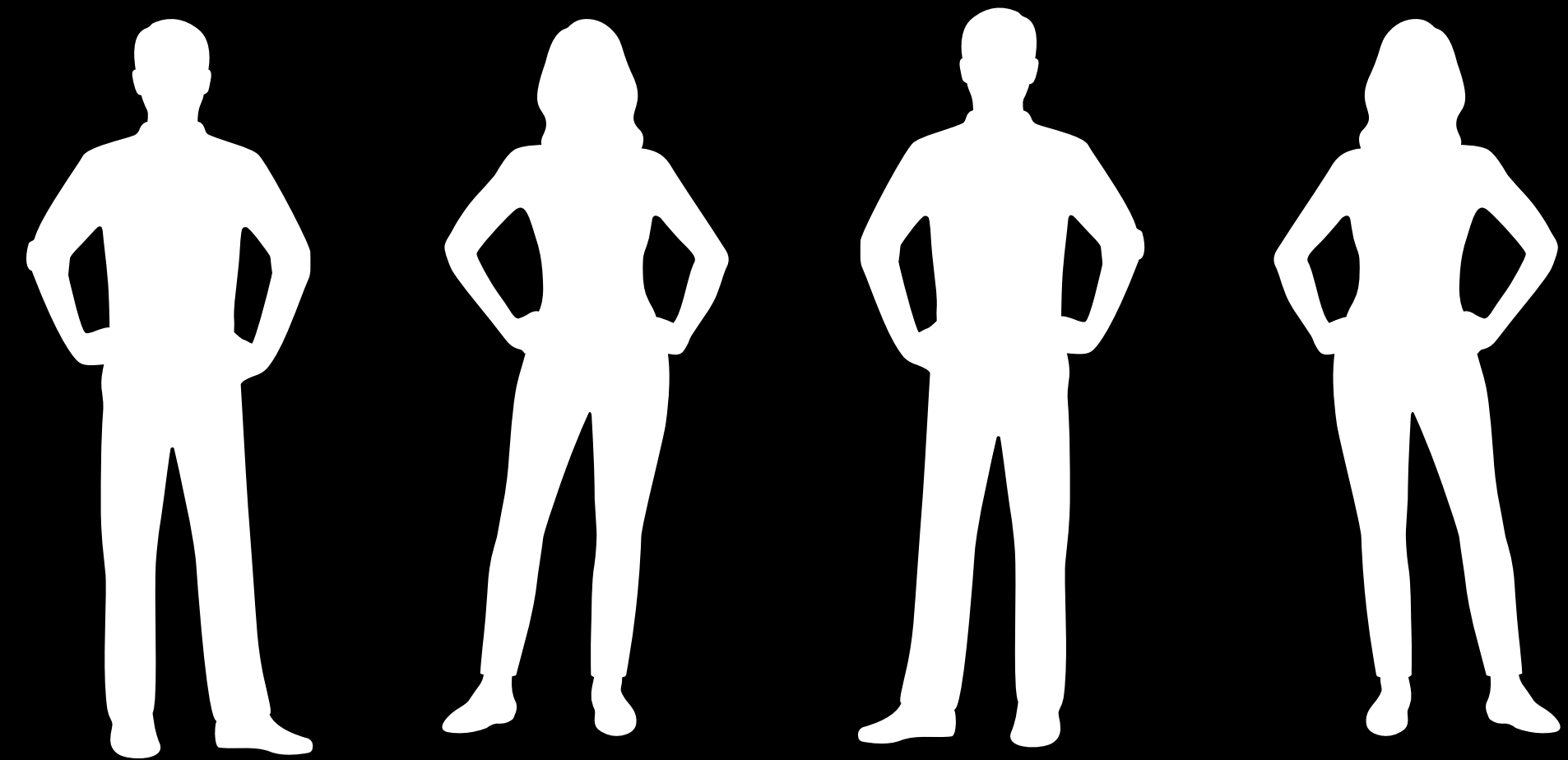
SOCIAL DEBT

It is the additional costs of a software project that results from unpredicted and unmanaged **socio-technical anti-patterns** in the software community [1].



SOCIAL DEBT

It is the additional costs of a software project that results from unpredicted and unmanaged **socio-technical anti-patterns** in the software community [1].



Socio-Technical

WHAT
DO YOU
MEAN
?

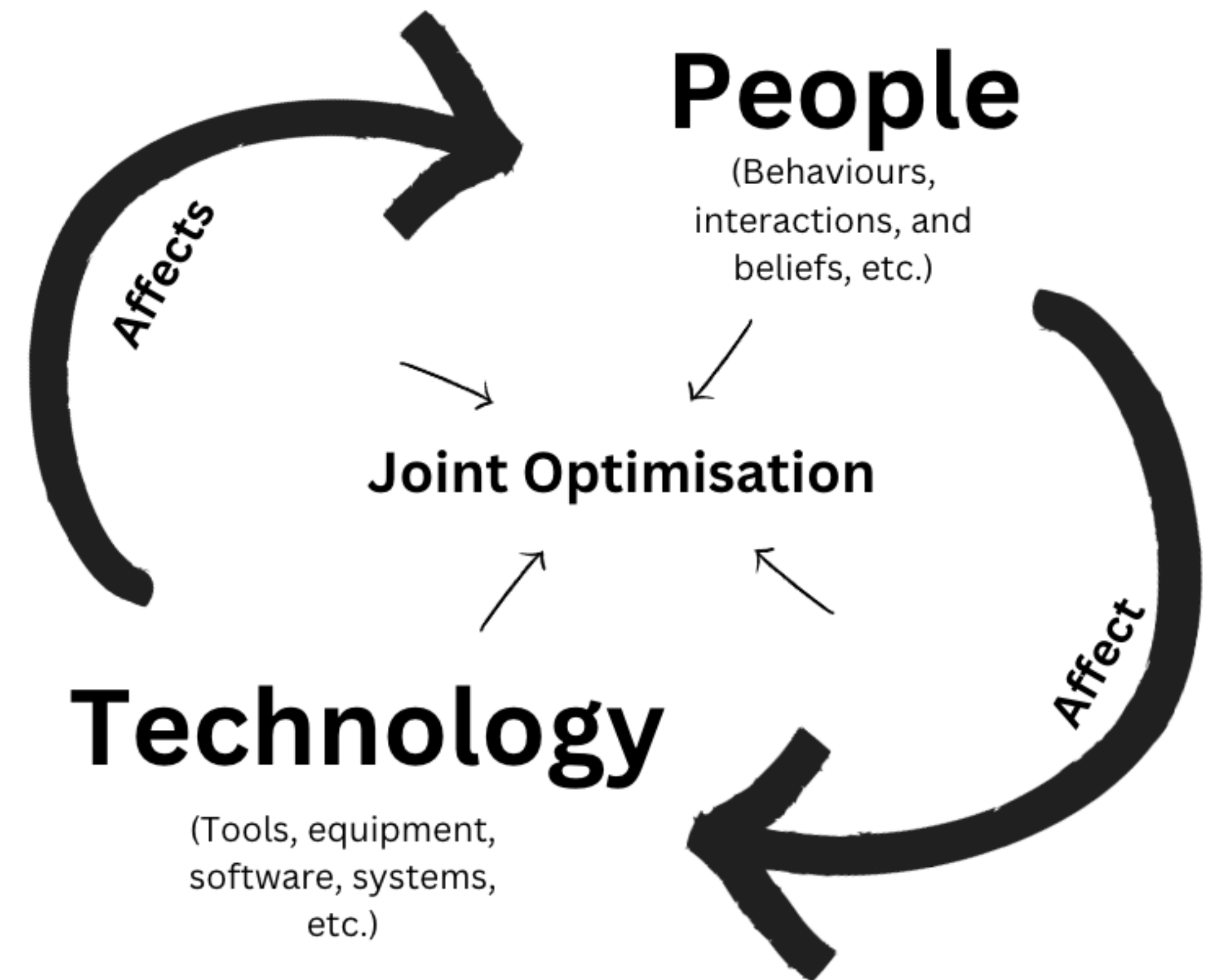
Socio-Technical

What does it mean?

In software development, technology is an integral part of stakeholder life.

Studying the interactions of these stakeholders without considering the technological aspect is likely to lead to false or incomplete results, which are thus **useless** [2].

Sociotechnical Theory



Socio-Technical

Anti-patterns

WHAT
DO YOU
MEAN
?

Socio-Technical

Anti-patterns

What does it mean?

An anti-pattern is a common response to a recurring problem that is usually ineffective and risks being highly counterproductive.



Social

Anti-patterns

Socio-Technical

Anti-patterns

What is the difference?

Social anti-patterns refer to negative or counterproductive behaviors and practices within social systems, particularly in organizational or team settings.

Socio-technical anti-patterns refer to negative or counterproductive behaviors and practices within systems in which people and technology are highly connected

SOCIAL DEBT

How can we manage it?





You can't **control**
what you can't
measure

— Tom DeMarco



Code Smells

Code Smells are symptoms of poor design or implementation choices

– Marting Fowler

```
public abstract class AbstractCollection implements Collection {  
    public void addAll(AbstractCollection c) {  
        if (c instanceof Set) {  
            Set s = (Set)c;  
            for (int i=0; i < s.size(); i++) {  
                if (!contains(s.getElementAt(i))) {  
                    add(s.getElementAt(i));  
                }  
            }  
        }  
        else if (c instanceof List) {  
            List l = (List)c;  
            for (int i=0; i < l.size(); i++) {  
                if (!contains(l.get(i))) {  
                    add(l.get(i));  
                }  
            }  
        }  
        else if (c instanceof Map) {  
            Map m = (Map)c;  
            for (int i=0; i<m.size(); i++)  
                add(m.keys[i], m.values[i]);  
        }  
    }  
}
```

Duplicated Code

Duplicated Code

Alternative Classes with Different Interfaces

Switch Statement

Inappropriate Intimacy

Long Method

~~Code~~ Community Smells

Community Smells are socio-technical anti-patterns in the communication and collaboration structure of a software community [3].



~~Code~~ Community Smells

Community Smells are socio-technical anti-patterns in the communication and collaboration structure of a software community [3].

Code Co Smells

Community Smells are
socio-technical anti-
patterns in the communication
collaboration structure of
software community

Community Smells - The Sources of Social Debt: A Systematic Literature Review

Eduardo Caballero-Espinosa^{a,b}, Jeffrey C. Carver^c, Kimberly Stowers^d

^a*Center for Research, Development, and Innovation in Information and Communication
Technology (CIDITIC) - Technological University of Panama*

^b*Computing Systems Engineering Department - Technological University of Panama*

^c*Department of Computer Science, The University of Alabama, Tuscaloosa, AL, USA*

^d*Tim Fletcher, Co., San Jose, CA, USA*

Abstract

Context: Social debt describes the accumulation of unforeseen project costs (or potential costs) from sub-optimal software development processes. Community smells are sociotechnical anti-patterns and one source of social debt. Because community smells impact software teams, development processes, outcomes, and organizations, we to understand their impact on software engineering.

Objective: To provide an overview of community smells in social debt, based on published literature, and describe future research.

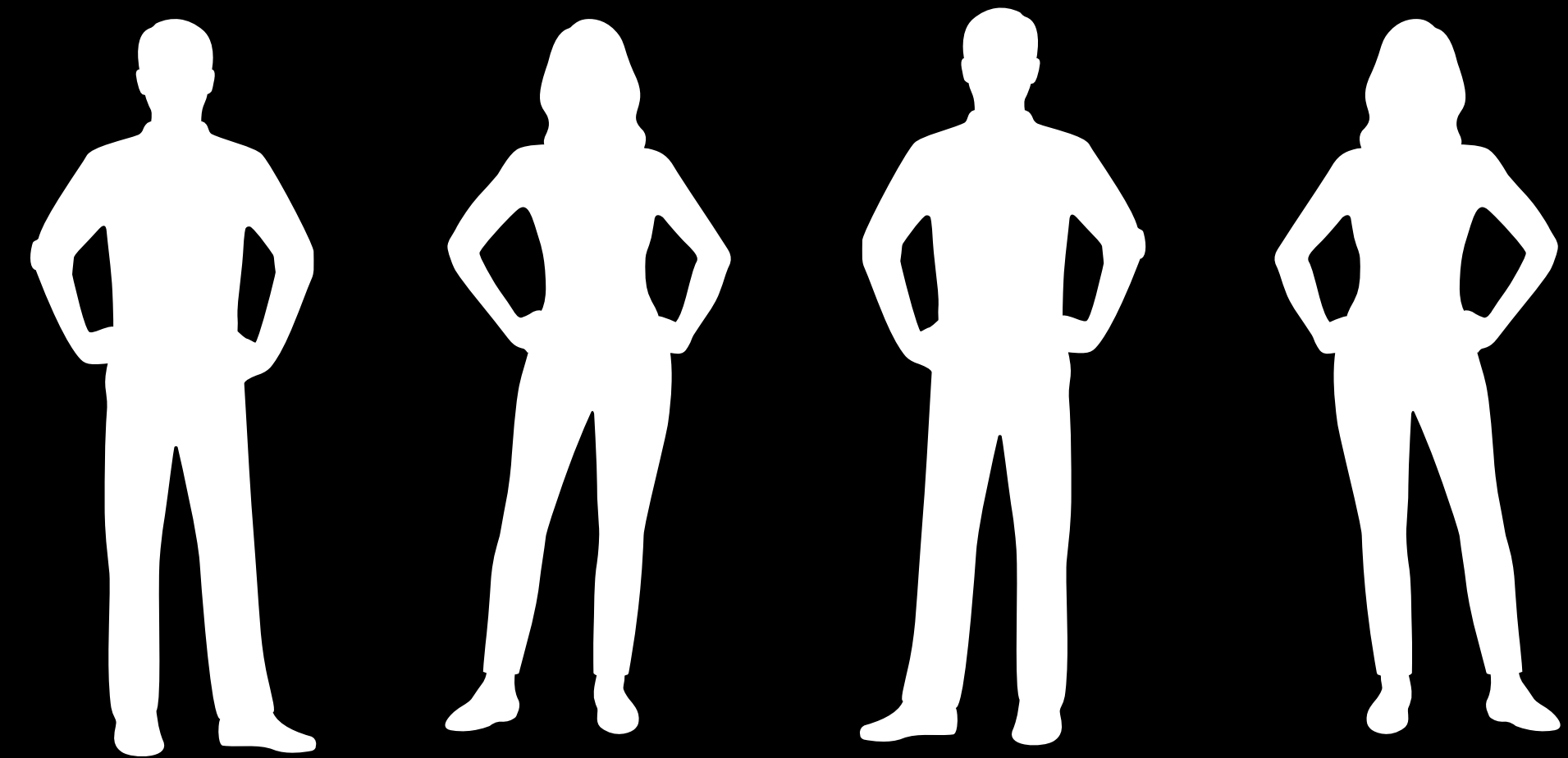
Method: We conducted a systematic literature review (SLR) to identify properties, understand origins and evolution, and describe the emergence of community smells. This SLR explains the impact of community smells on teamwork and team performance.

Results: We include 25 studies. Social debt describes the impacts of poor socio-technical decisions on work environments, people, software products, and society. For each of the 30 community smells identified as sources of social debt, we provide a detailed description, management approaches, organizational strategies, and mitigation effectiveness. We identify five groups of management approaches: organizational strategies, frameworks, models, tools, and guidelines. We describe 11 common properties of community smells. We develop the *Community Smell Stages Framework* to concisely

SOCIAL DEBT

It is the additional costs of a software project that results from unpredicted and unmanaged **socio-technical anti-patterns** in the software community [1].

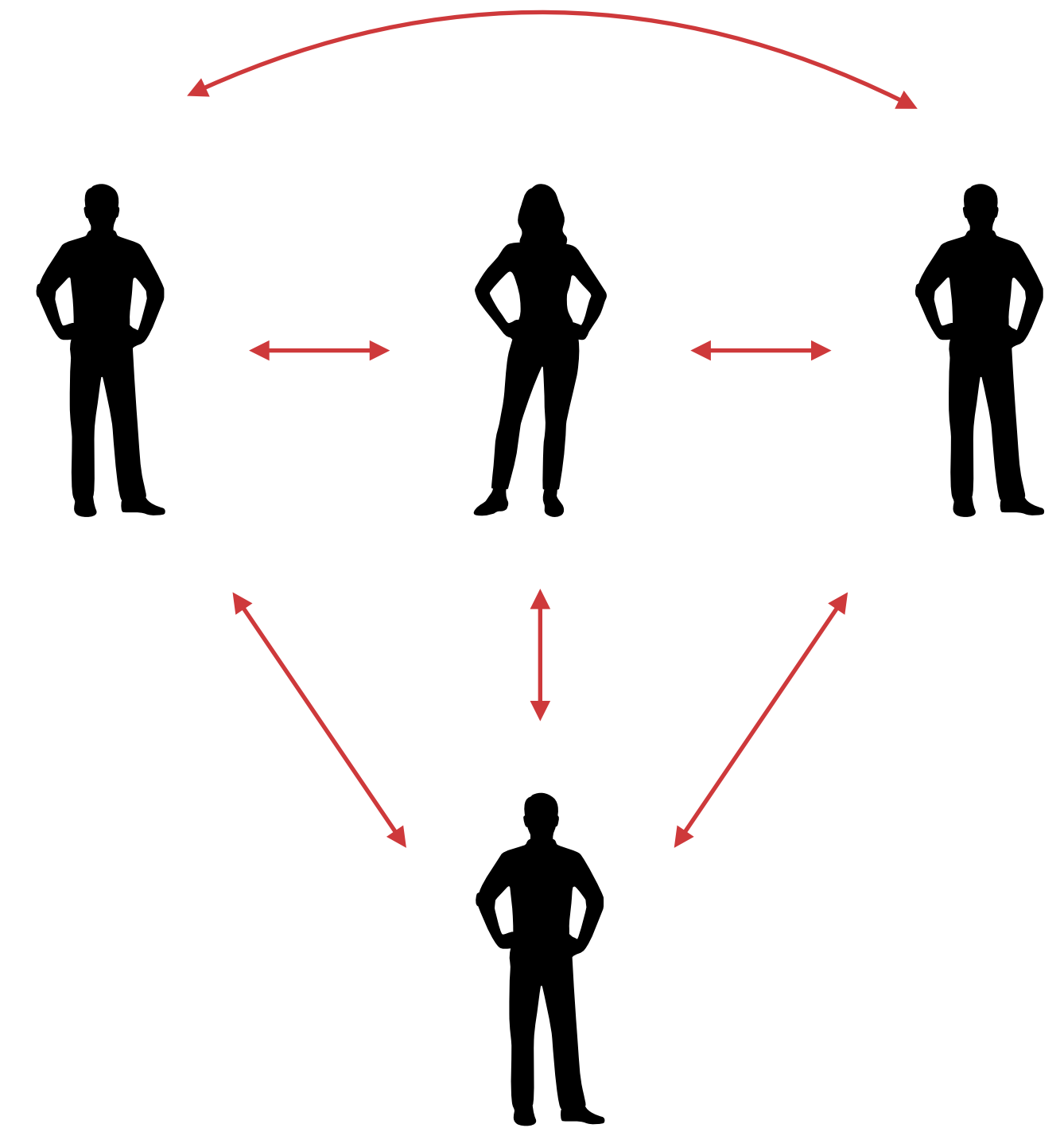
Community smells—i.e., **socio-technical anti-patterns** in the development community—are precursors of social debt.



Black Cloud

A Community Smell Example

This smell is present when organizations do not provide the conditions for social interactions and effective communication between teammates. Thus, the conditions do not support the exchange of knowledge during software development processes, e.g., professional experience or understanding of projects in progress [3].

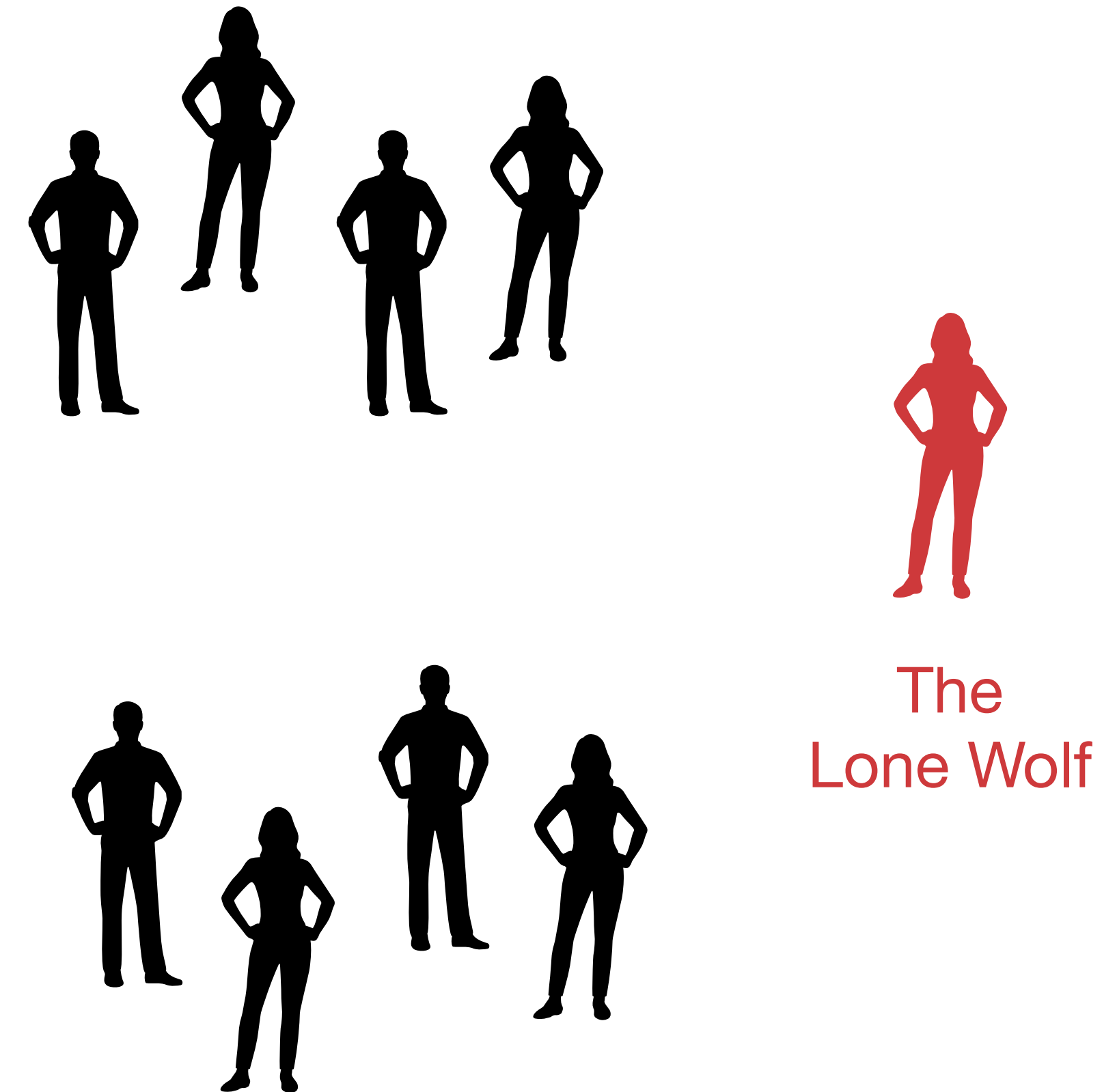


Lone Wolf

A Community Smell Example

is a scenario in which a team member isolates himself and works alone and without communicating with his co-workers.

This situation leads to tasks performed in isolation and loss of project tracking, as well as general discontent in the group [3].

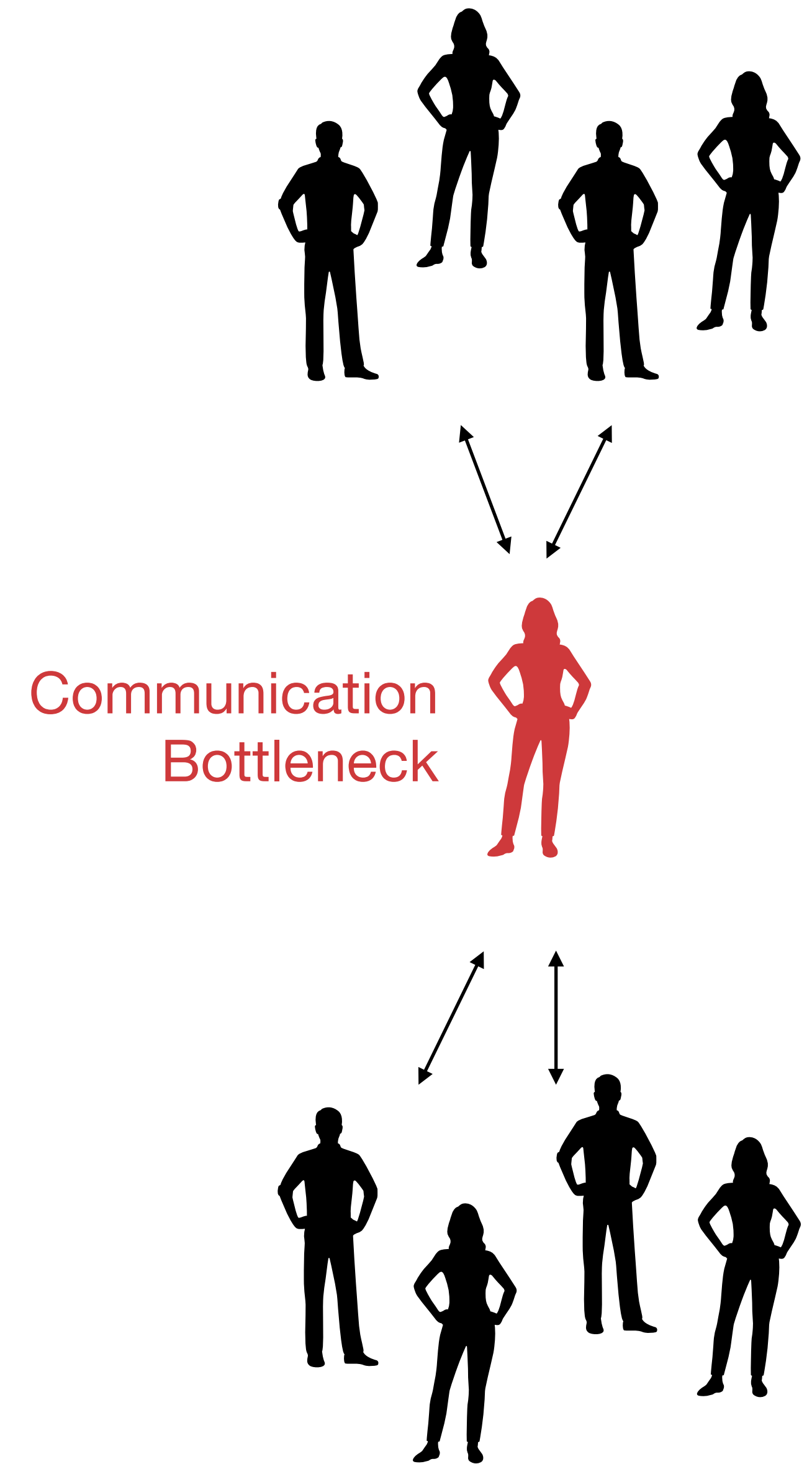


Radio Silence

A Community Smell Example

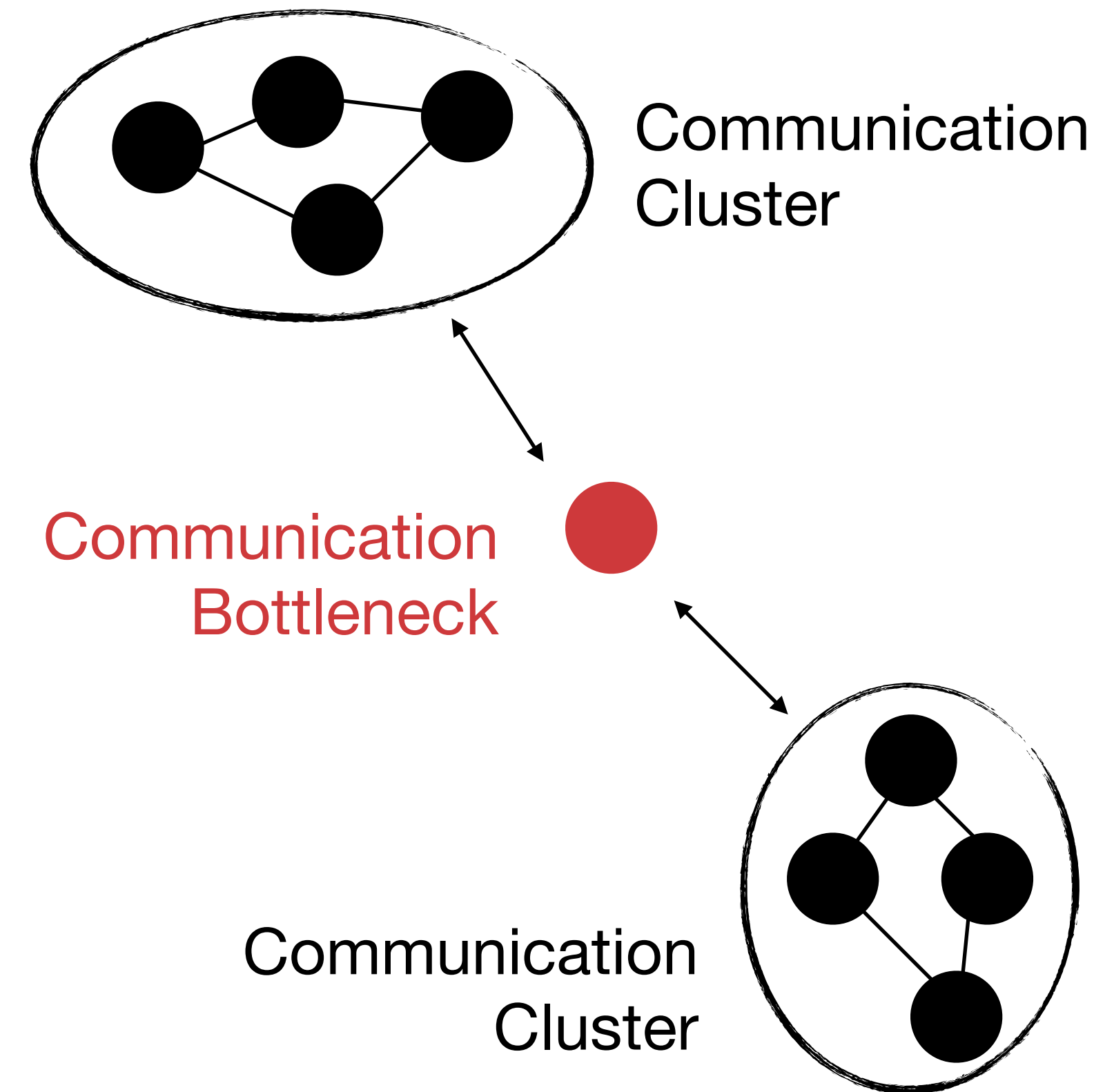
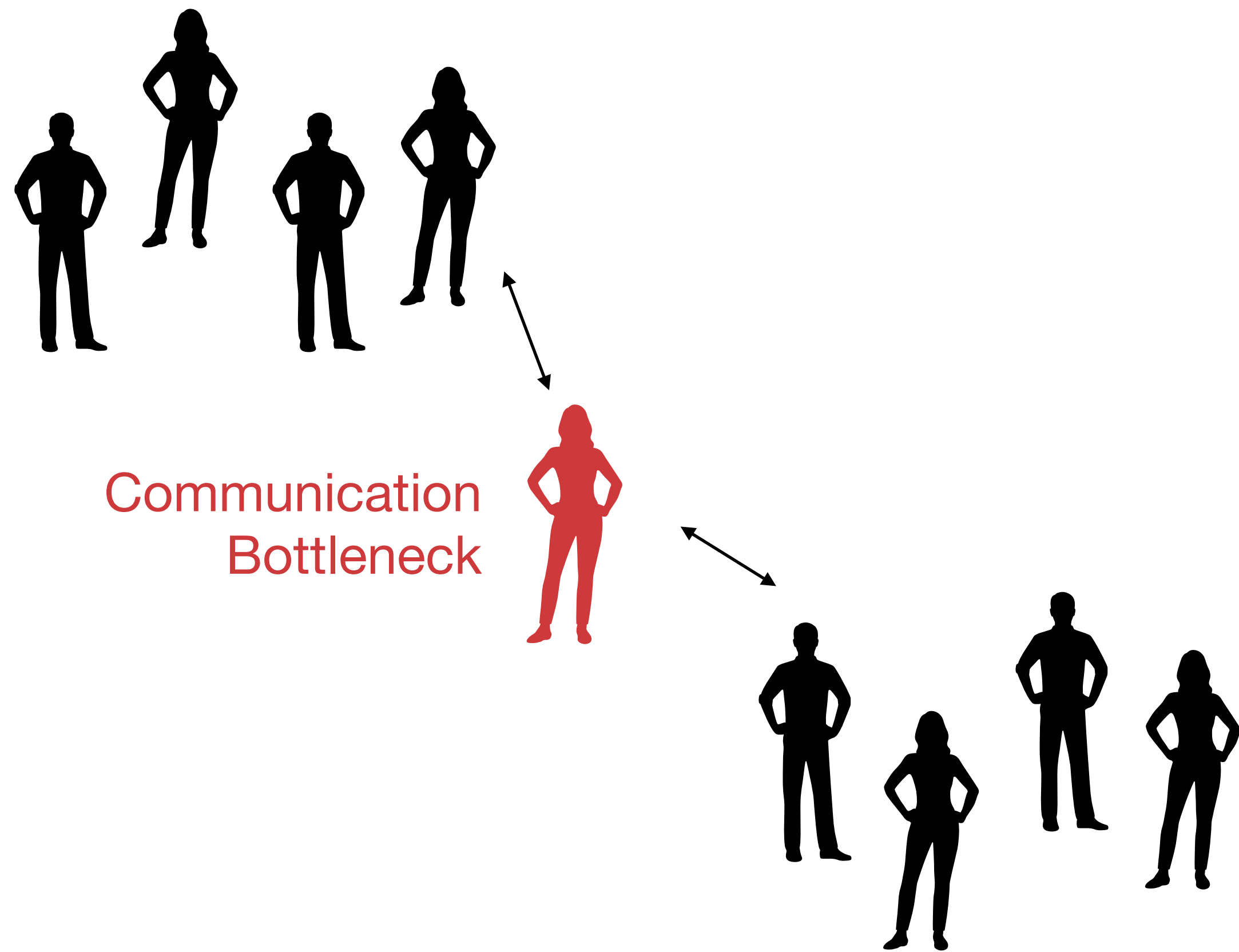
It is a scenario where leaders and teammates perform tasks in very formal and complex organizations. Under these conditions, team communication structures are not conducive to efficiently spreading information across the teams.

For instance, a person working as a unique information intermediary for different teams leads to communication overload and massive delays [3].

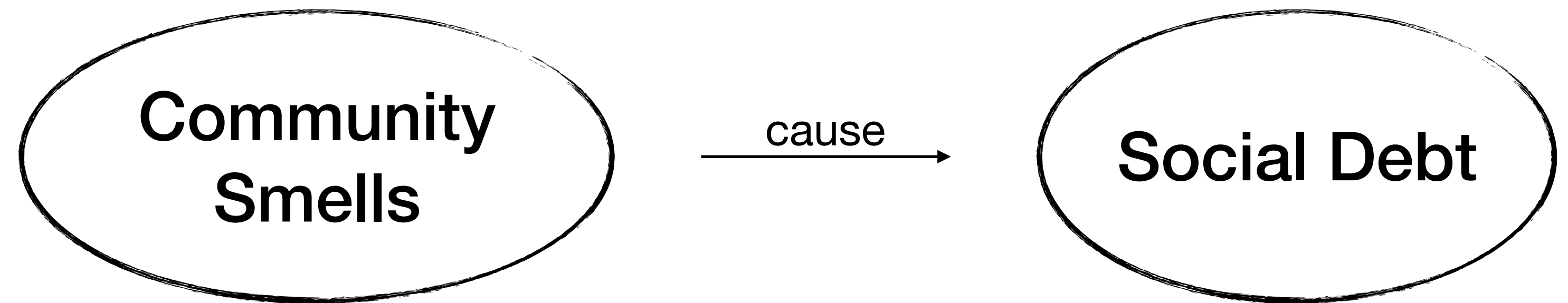


Radio Silence

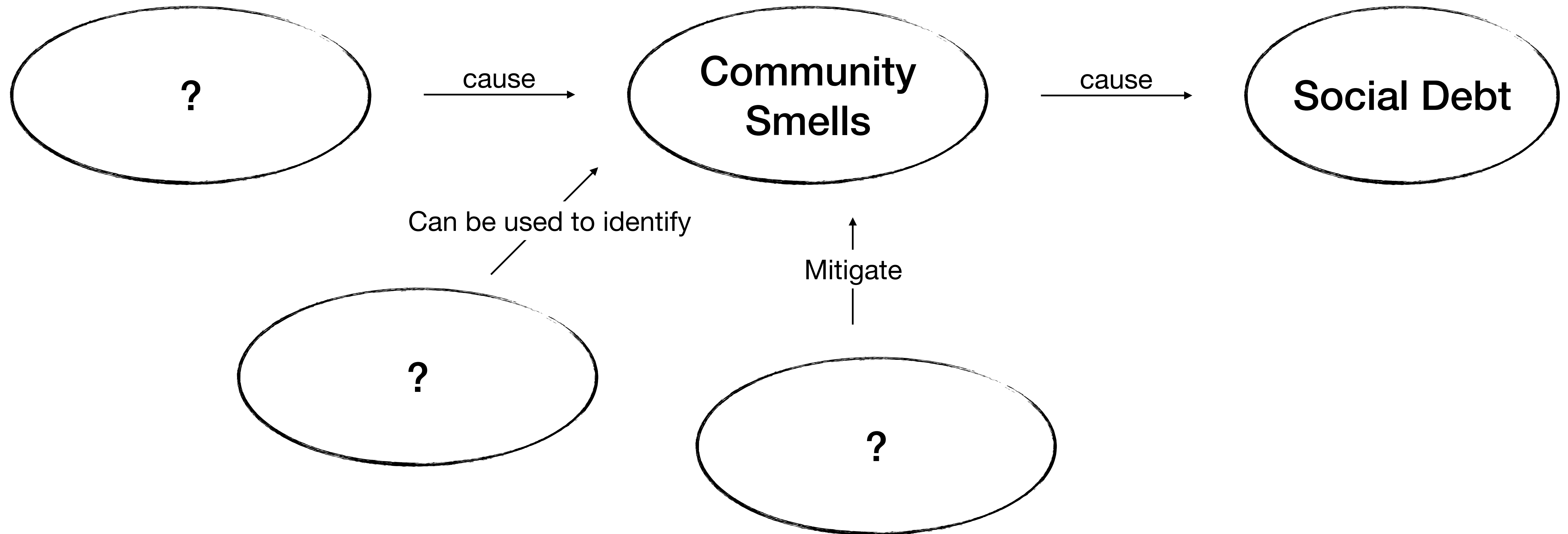
Representing Smells as graphs



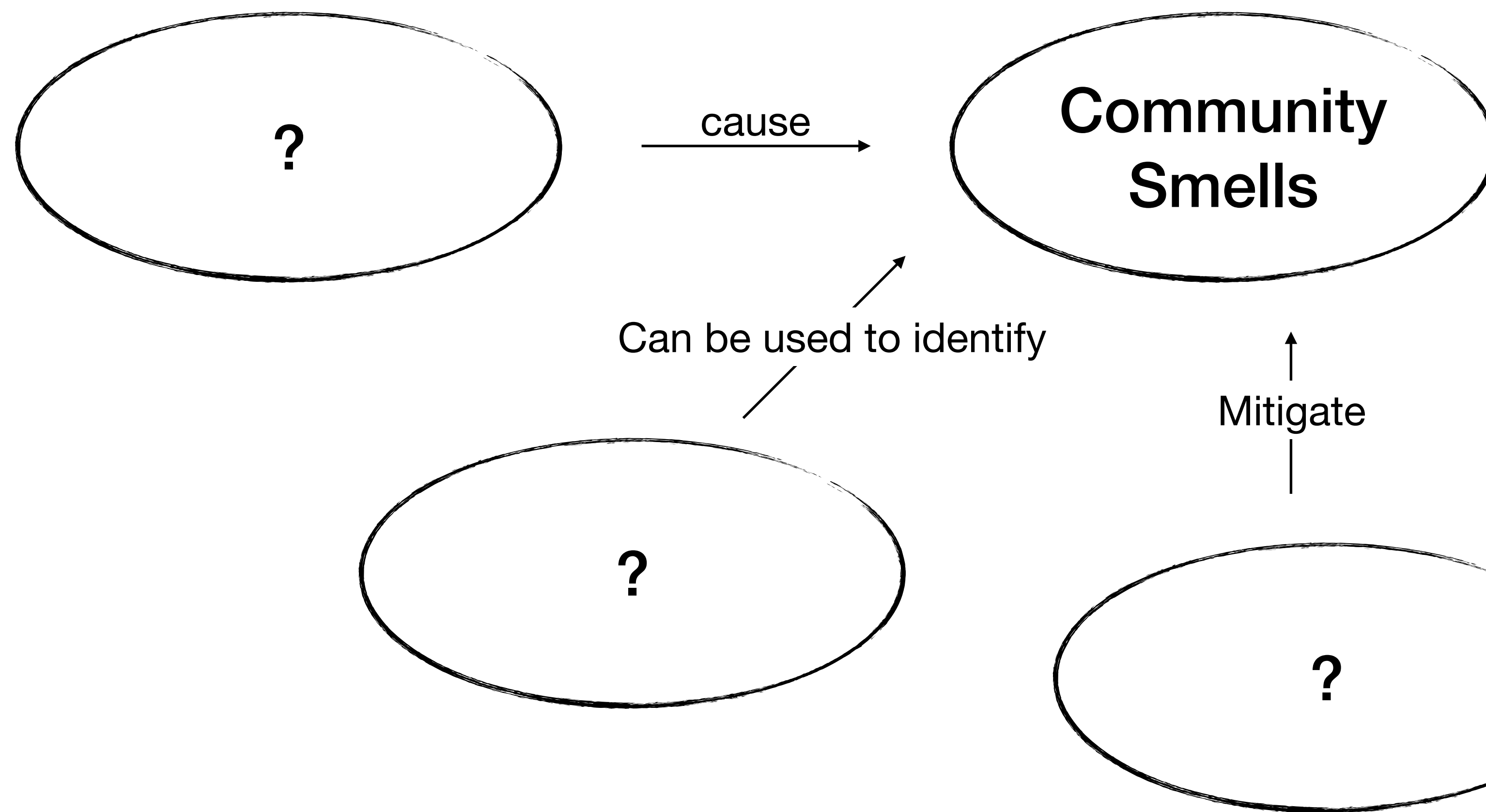
Community Smells & Social Debt



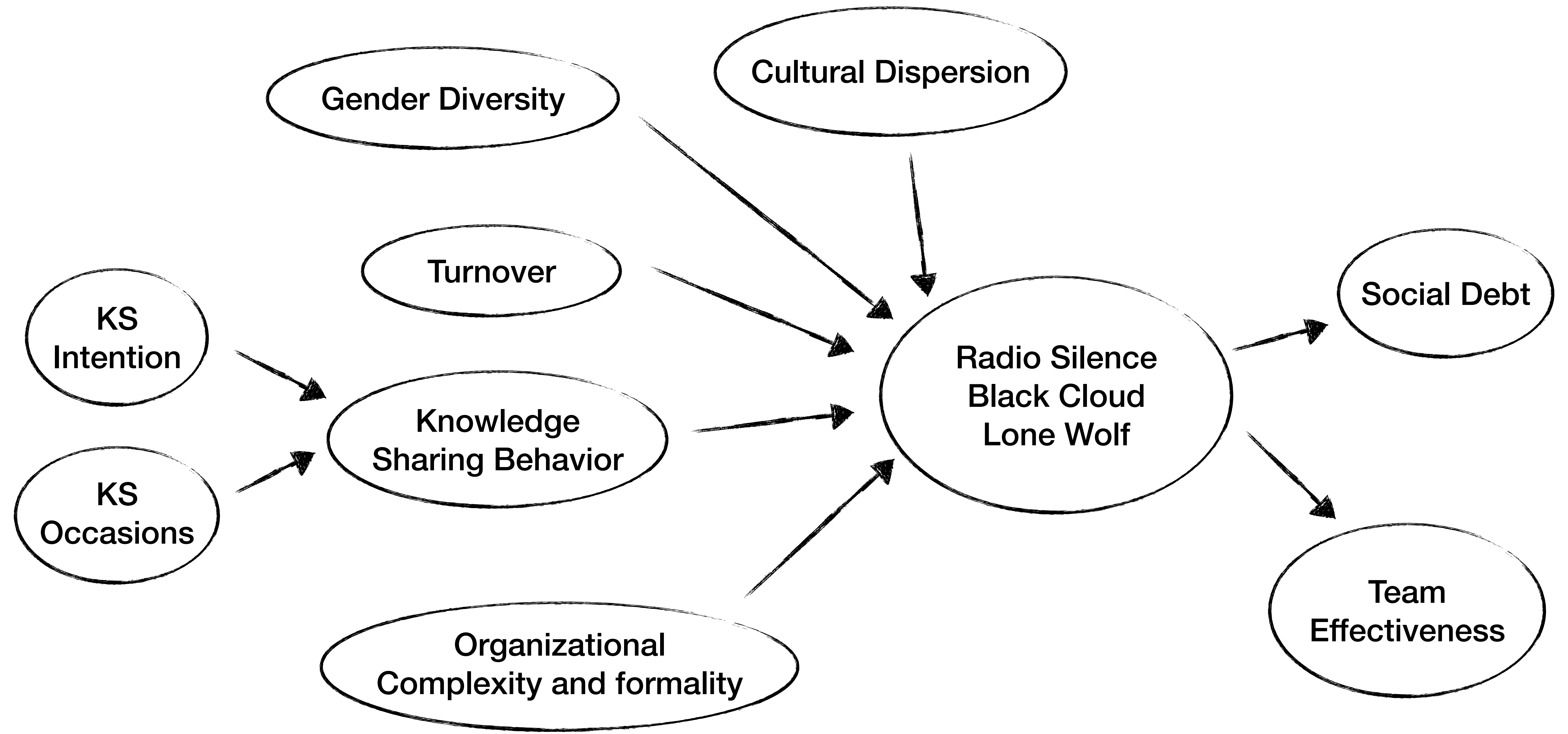
Community Smells & Social Debt



Research on Community Smells



Factors Related to Community Smells – a **Research** Perspective



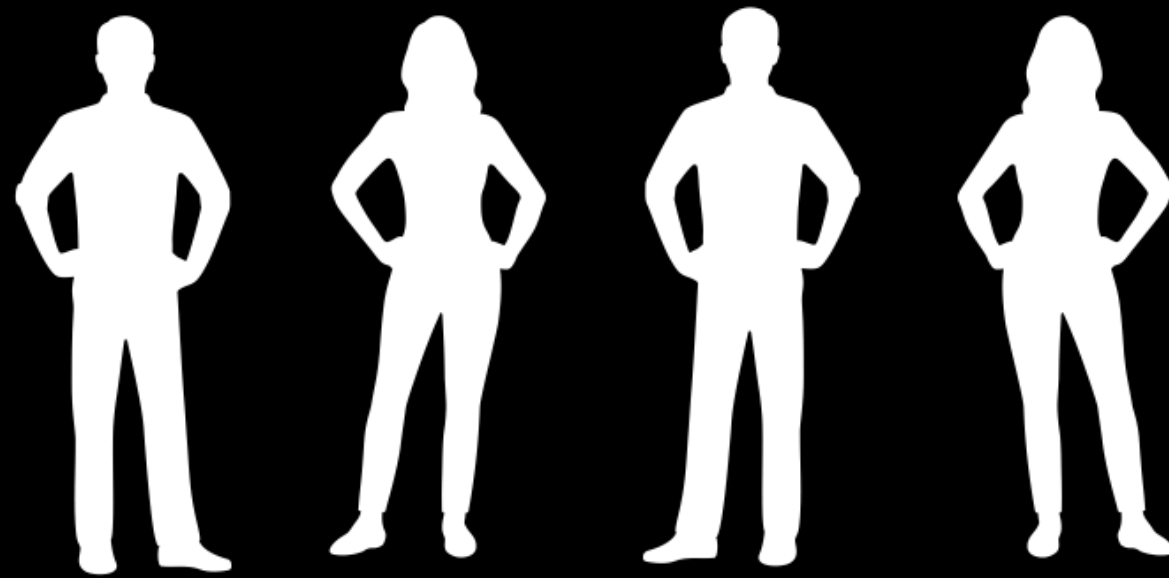
Conclusions

What do you (should) have learned?

SOCIAL DEBT

It is the additional costs of a software project that results from unpredicted and unmanaged **socio-technical anti-patterns** in the software community [1].

Community smells—i.e., **socio-technical anti-patterns** in the development community—are precursors of social debt.



- What is Social Debt
- What is a socio-technical anti-pattern
- What are community smells
- How are community smells related to social debt

[1] Tamburri, Damian A., et al. "What is social debt in software engineering?." 2013 6th International Workshop on Cooperative and Human Aspects of Software Engineering (CHASE). IEEE, 2013.

External Research Material

- [1] Tamburri, Damian A., et al. "What is social debt in software engineering?." 2013 6th International Workshop on Cooperative and Human Aspects of Software Engineering (CHASE). IEEE, 2013.
- [2] Hoda, Rashina. "Socio-technical grounded theory for software engineering." IEEE Transactions on Software Engineering 48.10 (2021): 3808-3832.
- [3] Caballero-Espinosa, Eduardo, et al. "Community smells—The sources of social debt: A systematic literature review." Information and Software Technology 153 (2023): 107078.
- [4] Palomba, Fabio, et al. "Beyond technical aspects: How do community smells influence the intensity of code smells?." IEEE transactions on software engineering 47.1 (2018): 108-129.
- [5] Catolino, Gemma, et al. "Gender diversity and women in software teams: How do they affect community smells?." 2019 IEEE/ACM 41st International Conference on Software Engineering: Software Engineering in Society (ICSE-SEIS). IEEE, 2019.
- [6] Lambiase, Stefano, et al. "Good fences make good neighbours? on the impact of cultural and geographical dispersion on community smells." Proceedings of the 2022 ACM/IEEE 44th International Conference on Software Engineering: Software Engineering in Society. 2022.
- [7] Almarimi, Nuri, Ali Ouni, and Mohamed Wiem Mkaouer. "Learning to detect community smells in open source software projects." Knowledge-Based Systems 204 (2020): 106201.